CPU Scheduling: Student Assignment Worksheet

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Instructions for all problems

1. Draw a Gantt chart, including any CPU idle periods.
2. Find each process's completion time (CT).
3. Calculate each process's turnaround time (TAT) and waiting time (WT).
4. Calculate average turnaround time and average waiting time.

Formulas

TAT = CT − AT

WT = TAT − BT

Average TAT = Sum of TAT values ÷ Number of processes

Average WT = Sum of WT values ÷ Number of processes

Assumptions and tie-breaking rules

- All times are in milliseconds (ms). There is one CPU, no I/O waiting, and zero context-switch overhead.
- If processes arrive simultaneously, use their listed order.
- For equal burst times, remaining times, or priorities, prefer the earlier-arriving process, then the listed order.
- Do not preempt a running process merely because another process has equal remaining time or equal priority.
- Smaller priority numbers mean higher priority.
- In Round Robin, arrivals at the exact end of a time quantum enter the queue before the unfinished running process is requeued.

Problem 1: FCFS — All Processes Arrive Together

Schedule the following processes using First Come, First Served.

Process	Arrival time (ms)	Burst time (ms)
P1	0	6
P2	0	3
P3	0	8
P4	0	2

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Which process has the longest waiting time?

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

_____

Problem 2: FCFS — Different Arrival Times and CPU Idle Time

Schedule the following processes using FCFS.

Process	Arrival time (ms)	Burst time (ms)
P1	0	3
P2	5	4
P3	6	2
P4	12	3

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Mark every CPU idle interval.
- Calculate the total idle time between time 0 and the completion of all processes.

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

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Problem 3: SJF — All Processes Arrive Together

Schedule the following processes using non-preemptive Shortest Job First.

Process	Arrival time (ms)	Burst time (ms)
P1	0	7
P2	0	4
P3	0	1
P4	0	5
P5	0	2

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Repeat the scheduling using FCFS. Which algorithm gives a lower average waiting time for this workload?

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			
P5			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

_____

Problem 4: SJF — Different Arrival Times

Schedule the following processes using non-preemptive SJF.

Process	Arrival time (ms)	Burst time (ms)
P1	0	8
P2	1	4
P3	2	2
P4	3	1
P5	10	3

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Which process runs immediately after P1?
- Why does P4 not interrupt P1 when it arrives?

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			
P5			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

_____

Problem 5: SRTF — Preemptive Scheduling

Schedule the following processes using Shortest Remaining Time First.

Process	Arrival time (ms)	Burst time (ms)
P1	0	9
P2	1	5
P3	2	2
P4	4	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Identify every preemption: a running process being interrupted before completing.
- At time 4, which process should run next, and why?

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

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Problem 6: Round Robin — Time Quantum of 2 ms

Schedule the following processes using Round Robin with a time quantum of 2 ms.

Process	Arrival time (ms)	Burst time (ms)
P1	0	5
P2	0	4
P3	0	2
P4	0	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Write the ready-queue order immediately after each time slice.
- Which process finishes first?

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

_____

Problem 7: Round Robin — Processes Arrive During Execution

Schedule the following processes using Round Robin with a time quantum of 3 ms.

Process	Arrival time (ms)	Burst time (ms)
P1	0	7
P2	1	4
P3	3	2
P4	5	5

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Show the ready-queue order immediately after P1's first time slice.
- Explain where P3 is placed when it arrives at time 3. Apply the boundary-arrival rule.

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

_____

Problem 8: Non-Preemptive Priority Scheduling

Schedule the following processes using non-preemptive priority scheduling.

Process	Arrival time (ms)	Burst time (ms)	Priority
P1	0	5	3
P2	1	3	1
P3	2	4	4
P4	3	2	2
P5	4	1	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Why does P1 continue running after P2 arrives?
- When the CPU becomes free, how is the tie between P2 and P5 resolved?

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			
P5			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

_____

Problem 9: Preemptive Priority Scheduling

Schedule the following processes using preemptive priority scheduling.

Process	Arrival time (ms)	Burst time (ms)	Priority
P1	0	8	4
P2	1	4	2
P3	2	2	1
P4	3	3	3
P5	5	1	1

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Identify every preemption.
- Which process has the longest waiting time?

Gantt chart

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Calculations

Process	CT (ms)	TAT (ms)	WT (ms)
P1			
P2			
P3			
P4			
P5			

Average TAT: _____ms Average WT: _____ms

Additional answers / working:

_____

_____

Problem 10: Compare Four Scheduling Algorithms

Solve using FCFS, non-preemptive SJF, SRTF, and Round Robin with a time quantum of 2 ms.

Process	Arrival time (ms)	Burst time (ms)
P1	0	6
P2	1	3
P3	2	1
P4	3	4

Complete all tasks from the general instructions, then answer the additional questions.

Additional questions

- Which algorithm gives the lowest average waiting time for this workload?
- Does that algorithm also give the lowest average turnaround time?
- Do any algorithms produce the same execution order for this particular process set?

Algorithm	Average TAT (ms)	Average WT (ms)
FCFS		
SJF		
SRTF		
Round Robin (q = 2 ms)		

Working space

Use an additional sheet for the four Gantt charts and process calculations.

PROBLEM 1: FCFS — ALL PROCESSES ARRIVE TOGETHER

Gantt Chart

0 | P1 | 6 | P2 | 9 | P3 | 17 | P4 | 19

Calculations

Process	CT	TAT = CT − AT	WT = TAT − BT
P1	6	6	0
P2	9	9	6
P3	17	17	9
P4	19	19	17

Average TAT = (6 + 9 + 17 + 19) / 4 = 12.75 ms

Average WT = (0 + 6 + 9 + 17) / 4 = 8 ms

Additional answer: P4 has the longest waiting time of 17 ms.

PROBLEM 2: FCFS — DIFFERENT ARRIVAL TIMES AND CPU IDLE TIME

Gantt Chart

0 | P1 | 3 | IDLE | 5 | P2 | 9 | P3 | 11 | IDLE | 12 | P4 | 15

Calculations

Process	CT	TAT	WT
P1	3	3	0
P2	9	4	0
P3	11	5	3
P4	15	3	0

Average TAT = (3 + 4 + 5 + 3) / 4 = 3.75 ms

Average WT = (0 + 0 + 3 + 0) / 4 = 0.75 ms

Additional answers:

1. First idle interval: 3–5 ms.

2. Second idle interval: 11–12 ms.

3. Total CPU idle time = 2 + 1 = 3 ms.

PROBLEM 3: NON-PREEMPTIVE SJF — ALL PROCESSES ARRIVE TOGETHER

Gantt Chart (SJF)

0 | P3 | 1 | P5 | 3 | P2 | 7 | P4 | 12 | P1 | 19

Calculations

Process	CT	TAT	WT
P1	19	19	12
P2	7	7	3
P3	1	1	0
P4	12	12	7
P5	3	3	1

Average TAT = (19 + 7 + 1 + 12 + 3) / 5 = 8.4 ms

Average WT = (12 + 3 + 0 + 7 + 1) / 5 = 4.6 ms

Repeat using FCFS

FCFS Gantt Chart:

0 | P1 | 7 | P2 | 11 | P3 | 12 | P4 | 17 | P5 | 19

Process	CT	TAT	WT
P1	7	7	0
P2	11	11	7
P3	12	12	11
P4	17	17	12
P5	19	19	17

Average TAT = 66 / 5 = 13.2 ms

Average WT = 47 / 5 = 9.4 ms

Additional answer: SJF gives the lower average waiting time (4.6 ms), compared with FCFS (9.4 ms).

PROBLEM 4: NON-PREEMPTIVE SJF — DIFFERENT ARRIVAL TIMES

Gantt Chart

0 | P1 | 8 | P4 | 9 | P3 | 11 | P5 | 14 | P2 | 18

Calculations

Process	CT	TAT	WT
P1	8	8	0
P2	18	17	13
P3	11	9	7
P4	9	6	5
P5	14	4	1

Average TAT = (8 + 17 + 9 + 6 + 4) / 5 = 8.8 ms

Average WT = (0 + 13 + 7 + 5 + 1) / 5 = 5.2 ms

Additional answers:

1. P4 runs immediately after P1.
2. P4 does not interrupt P1 because SJF is non-preemptive. Once P1 starts executing, it continues until completion, even when a shorter process arrives.

PROBLEM 5: SRTF — PREEMPTIVE SCHEDULING

Gantt Chart

0 | P1 | 1 | P2 | 2 | P3 | 4 | P4 | 5 | P2 | 9 | P1 | 17

Calculations

Process	CT	TAT	WT
P1	17	17	8
P2	9	8	3
P3	4	2	0
P4	5	1	0

Average TAT = (17 + 8 + 2 + 1) / 4 = 7 ms

Average WT = (8 + 3 + 0 + 0) / 4 = 2.75 ms

Additional answers:

1. At time 1 ms, P1 is interrupted by P2 because P2 has a shorter remaining time.
2. At time 2 ms, P2 is interrupted by P3 because P3 has the shortest remaining time.

3. At time 4 ms, P3 completes. P4 runs next because its burst time is 1 ms, which is the shortest among the available processes.

PROBLEM 6: ROUND ROBIN — TIME QUANTUM OF 2 MS

Gantt Chart

0 | P1 | 2 | P2 | 4 | P3 | 6 | P4 | 7 | P1 | 9 | P2 | 11 | P1 | 12

Calculations

Process	CT	TAT	WT
P1	12	12	7
P2	11	11	7
P3	6	6	4
P4	7	7	6

Average TAT = (12 + 11 + 6 + 7) / 4 = 9 ms

Average WT = (7 + 7 + 4 + 6) / 4 = 6 ms

Ready-Queue Order

- After P1's first slice (0–2): P2, P3, P4, P1
- After P2's first slice (2–4): P3, P4, P1, P2
- After P3 finishes (4–6): P4, P1, P2
- After P4 finishes (6–7): P1, P2
- After P1's second slice (7–9): P2, P1
- After P2 finishes (9–11): P1

Additional answer: P4 finishes first, at 7 ms.

PROBLEM 7: ROUND ROBIN — PROCESSES ARRIVE DURING EXECUTION

Time quantum = 3 ms.

Gantt Chart

0 | P1 | 3 | P2 | 6 | P3 | 8 | P1 | 11 | P4 | 14 | P2 | 15 | P1 | 16 | P4 | 18

Calculations

Process	CT	TAT	WT
P1	16	16	9
P2	15	14	10
P3	8	5	3
P4	18	13	8

Average TAT = (16 + 14 + 5 + 13) / 4 = 12 ms

Average WT = (9 + 10 + 3 + 8) / 4 = 7.5 ms

Additional answers:

1. Ready queue after P1's first time slice: P2, P3, P1.
2. P3 arrives exactly at time 3 ms, when P1's time quantum ends. According to the given boundary-arrival rule, P3 enters the ready queue before the unfinished P1 is added back to the queue.

PROBLEM 8: NON-PREEMPTIVE PRIORITY SCHEDULING

Smaller priority numbers indicate higher priority.

Gantt Chart

0 | P1 | 5 | P2 | 8 | P5 | 9 | P4 | 11 | P3 | 15

Calculations

Process	CT	TAT	WT
P1	5	5	0
P2	8	7	4
P3	15	13	9
P4	11	8	6
P5	9	5	4

Average TAT = (5 + 7 + 13 + 8 + 5) / 5 = 7.6 ms

Average WT = (0 + 4 + 9 + 6 + 4) / 5 = 4.6 ms

Additional answers:

1. P1 continues after P2 arrives because priority scheduling is non-preemptive. The running process cannot

be interrupted.

2. When the CPU becomes free at time 5 ms, P2 and P5 both have priority 1. P2 is selected because it arrived earlier.

PROBLEM 9: PREEMPTIVE PRIORITY SCHEDULING

Smaller priority numbers indicate higher priority.

Gantt Chart

0 | P1 | 1 | P2 | 2 | P3 | 4 | P2 | 5 | P5 | 6 | P2 | 8 | P4 | 11 | P1 | 18

Calculations

Process	CT	TAT	WT
P1	18	18	10
P2	8	7	3
P3	4	2	0
P4	11	8	5
P5	6	1	0

Average TAT = (18 + 7 + 2 + 8 + 1) / 5 = 7.2 ms

Average WT = (10 + 3 + 0 + 5 + 0) / 5 = 3.6 ms

Additional answers:

1. At time 1 ms, P1 is interrupted by P2 because P2 has a higher priority.
2. At time 2 ms, P2 is interrupted by P3 because P3 has a higher priority.
3. At time 5 ms, P2 is interrupted by P5 because P5 has priority 1, which is higher than P2's priority 2.
4. P1 has the longest waiting time of 10 ms.

PROBLEM 10: COMPARISON OF FOUR SCHEDULING ALGORITHMS

Processes:

P1: AT = 0, BT = 6

P2: AT = 1, BT = 3

P3: AT = 2, BT = 1

P4: AT = 3, BT = 4

A. FCFS

Gantt Chart:

0 | P1 | 6 | P2 | 9 | P3 | 10 | P4 | 14

Process	CT	TAT	WT
P1	6	6	0
P2	9	8	5
P3	10	8	7
P4	14	11	7

Average TAT = (6 + 8 + 8 + 11) / 4 = 8.25 ms

Average WT = (0 + 5 + 7 + 7) / 4 = 4.75 ms

B. Non-preemptive SJF

Gantt Chart:

0 | P1 | 6 | P3 | 7 | P2 | 10 | P4 | 14

Process	CT	TAT	WT
P1	6	6	0
P2	10	9	6
P3	7	5	4
P4	14	11	7

Average TAT = (6 + 9 + 5 + 11) / 4 = 7.75 ms

Average WT = (0 + 6 + 4 + 7) / 4 = 4.25 ms

C. SRTF

Gantt Chart:

0 | P1 | 1 | P2 | 2 | P3 | 3 | P2 | 5 | P4 | 9 | P1 | 14

Process	CT	TAT	WT
P1	14	14	8
P2	5	4	1
P3	3	1	0
P4	9	6	2

Average TAT = (14 + 4 + 1 + 6) / 4 = 6.25 ms

Average WT = (8 + 1 + 0 + 2) / 4 = 2.75 ms

D. Round Robin (Time Quantum = 2 ms)

Gantt Chart:

0 | P1 | 2 | P2 | 4 | P3 | 5 | P1 | 7 | P4 | 9 | P2 | 10 | P1 | 12 | P4 | 14

Process	CT	TAT	WT
P1	12	12	6
P2	10	9	6
P3	5	3	2
P4	14	11	7

Average TAT = (12 + 9 + 3 + 11) / 4 = 8.75 ms

Average WT = (6 + 6 + 2 + 7) / 4 = 5.25 ms

Final Comparison

Algorithm	Average TAT (ms)	Average WT (ms)
FCFS	8.25	4.75
SJF	7.75	4.25
SRTF	6.25	2.75
Round Robin (q = 2)	8.75	5.25

Additional answers:

1. SRTF gives the lowest average waiting time of 2.75 ms.
2. Yes. SRTF also gives the lowest average turnaround time of 6.25 ms.
3. No two algorithms produce the same complete execution order for this set of processes.

CONCLUSION

CPU scheduling algorithms decide the order in which processes use the CPU. FCFS follows arrival order, SJF selects the shortest available job, SRTF selects the shortest remaining job, Round Robin allocates fixed time slices, and priority scheduling selects processes according to priority. The calculations show that SRTF gives the lowest average waiting time and turnaround time for Problem 10.